Math 202: Ordinary Differential Equations

System of Damped Oscillators

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May 2026

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1 Abstract

This project investigates the motion of two coupled damped harmonic oscillators. The system consists of two masses connected by 3 springs, where damping causes energy loss over time. The study focuses on how the oscillators interact, how energy is transferred between them, and how the motion decays due to damping. The effect of changing parameters such as the damping coefficient and spring constant is studied to understand their influence on the system's behavior. We solved the two second-order differential equations using elimination and constant-coefficients methods, in addition to numerical solution and simulations using Python. The results show how the energy is continuously exchanged between the 2 masses through the springs, leading to coupled oscillations where the motion of one mass influences the other. Eventually, the system approaches equilibrium due to the damping effect.

2 Introduction

Coupled oscillatory systems are common in many areas of physics and engineering, including mechanical structures, vehicle suspension systems, and vibration studies. An oscillatory system is one in which a physical quantity, such as displacement, velocity, or energy, varies periodically in time around an equilibrium position due to restoring forces. When multiple oscillators are connected, their motions become interdependent, leading to more complex behavior such energy exchange between components.

In this project, we examine a model made up of two masses connected by 3 springs and influenced by damping forces. The motion of each mass depends not only on its own properties but also on its interaction with the other mass through the spring, resulting in a set of coupled differential equations. The goal of this work is to represent the system using second-order ordinary differential equations and investigate how key parameters such as mass, spring stiffness, and damping coefficients affect the system's behavior and explain the physical meaning of our solution. By simplifying the equations of the system and applying the constant-coefficient method, we aim to study the characteristics of the motion, including oscillatory behavior, energy transfer between the masses, and the gradual decay of energy caused by damping. Moreover, Numerical solution will be used for further explanation and understanding.

Overall, this project demonstrates how mathematical modeling can be used to represent real-world physical systems and provide insight into their dynamic responses.

3 Methodology

3.1 Assumptions

To model the system in a mathematical way, several simplifying assumptions are made:

- The motion of both masses is restricted to one dimension along a horizontal line.
- Both objects have equal masses
- The springs obey Hooke's law; it is ideal, massless, and exerts a linear restoring force proportional to displacement.
- The damping forces are linear and proportional to velocity.
- Energy is only dissipated through damping.
- The masses are treated as point particles, neglecting rotational effects and physical dimensions.
- External forces are neglected.
- All system parameters (mass, damping coefficients, and spring stiffness) are constant in time.

These assumptions allow the system to be modeled using second-order linear differential equations and easily solved using standard analytical and numerical methods.

3.1.1 Analytical Solution

To analyze the dynamics of the system, we first derive the governing equations of motion using Newton's second law.

Consider the damped coupled system

$$\{M\ddot{x}_1 + b\dot{x}_1 + (\kappa + \kappa_{12})x_1 - \kappa_{12}x_2 = 0 M\ddot{x}_2 + b\dot{x}_2 + (\kappa + \kappa_{12})x_2 - \kappa_{12}x_1 = 0 \tag{1}$$

where

- M is the mass,
- b is the damping coefficient,
- κ is the spring constant,
- κ_{12} is the coupling spring constant.

Step 1: Introduce Normal Coordinates

Define

$$u = x_1 + x_2 \tag{2}$$

$$v = x_1 - x_2 \tag{3}$$

Then

$$\dot{u} = \dot{x}_1 + \dot{x}_2 \tag{4}$$

$$\ddot{u} = \ddot{x}_1 + \ddot{x}_2 \tag{5}$$

and similarly

$$\dot{v} = \dot{x}_1 - \dot{x}_2 \tag{6}$$

$$\ddot{v} = \ddot{x}_1 - \ddot{x}_2 \tag{7}$$

Step 2: Add the Equations

Adding the original equations gives

$$M(\ddot{x}_1 + \ddot{x}_2) + b(\dot{x}_1 + \dot{x}_2) + (\kappa + \kappa_{12})(x_1 + x_2) - \kappa_{12}(x_1 + x_2) = 0 \tag{8}$$

Hence,

$$M\ddot{u} + b\dot{u} + \kappa u = 0 \tag{9}$$

This is the equation of a damped harmonic oscillator.

Step 3: Subtract the Equations

Subtracting the second equation from the first gives

$$M(\ddot{x}_1 - \ddot{x}_2) + b(\dot{x}_1 - \dot{x}_2) + (\kappa + \kappa_{12})(x_1 - x_2) + \kappa_{12}(x_1 - x_2) = 0 \tag{10}$$

Thus,

$$M\ddot{v} + b\dot{v} + (\kappa + 2\kappa_{12})v = 0 \tag{11}$$

Again, this is a damped harmonic oscillator.

Step 4: Solve for $u(t)$

Consider

$$M\ddot{u} + b\dot{u} + \kappa u = 0 \tag{12}$$

Assume a solution of the form

$$u = e^{rt} \tag{13}$$

Then

$$\dot{u} = re^{rt} \tag{14}$$

$$\ddot{u} = r^2 e^{rt} \tag{15}$$

Substituting into the equation:

$$Mr^2 e^{rt} + bre^{rt} + \kappa e^{rt} = 0 \tag{16}$$

Dividing by $e^{rt} \neq 0$,

$$Mr^2 + br + \kappa = 0 \tag{17}$$

Using the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4M\kappa}}{2M} \tag{18}$$

For the underdamped case,

$$b^2 < 4M\kappa \tag{19}$$

the roots are complex:

$$r = -\frac{b}{2M} \pm i\frac{\sqrt{4M\kappa - b^2}}{2M} \tag{20}$$

Define

$$\gamma = \frac{b}{2M} \tag{21}$$

and

$$\Omega_1 = \sqrt{\frac{\kappa}{M} - \gamma^2} \tag{22}$$

Hence,

$$r = -\gamma \pm i\Omega_1 \tag{23}$$

Therefore,

$$u(t) = e^{-\gamma t}\left[A\cos(\Omega_1 t) + B\sin(\Omega_1 t)\right] \tag{24}$$

Step 5: Solve for $v(t)$

Now consider

$$M\ddot{v} + b\dot{v} + (\kappa + 2\kappa_{12})v = 0 \tag{25}$$

Assume

$$v = e^{rt} \tag{26}$$

Then the characteristic equation becomes

$$Mr^2 + br + (\kappa + 2\kappa_{12}) = 0 \tag{27}$$

The roots are

$$r = \frac{-b \pm \sqrt{b^2 - 4M(\kappa + 2\kappa_{12})}}{2M} \tag{28}$$

For the underdamped case,

$$b^2 < 4M(\kappa + 2\kappa_{12}) \tag{29}$$

Define

$$\Omega_2 = \sqrt{\frac{\kappa + 2\kappa_{12}}{M} - \gamma^2} \tag{30}$$

Then

$$r = -\gamma \pm i\Omega_2 \tag{31}$$

Therefore,

$$v(t) = e^{-\gamma t}\left[C\cos(\Omega_2 t) + D\sin(\Omega_2 t)\right] \tag{32}$$

Step 6: Recover x_1 and x_2

Since

$$u = x_1 + x_2 \tag{33}$$

$$v = x_1 - x_2 \tag{34}$$

we solve for x_1 and x_2:

$$x_1 = \frac{u+v}{2} \tag{35}$$

$$x_2 = \frac{u-v}{2} \tag{36}$$

Substituting the expressions for u and v:

$$x_1(t) = \frac{1}{2}e^{-\gamma t}\left[A\cos(\Omega_1 t) + B\sin(\Omega_1 t) + C\cos(\Omega_2 t) + D\sin(\Omega_2 t)\right] \tag{37}$$

and

$$x_2(t) = \frac{1}{2}e^{-\gamma t}\left[A\cos(\Omega_1 t) + B\sin(\Omega_1 t) - C\cos(\Omega_2 t) - D\sin(\Omega_2 t)\right] \tag{38}$$

where

$$\gamma = \frac{b}{2M} \tag{39}$$

$$\Omega_1 = \sqrt{\frac{\kappa}{M} - \gamma^2} \tag{40}$$

and

$$\Omega_2 = \sqrt{\frac{\kappa + 2\kappa_{12}}{M} - \gamma^2} \tag{41}$$

Physical Interpretation

The motion is a superposition of two damped normal modes:

1. Symmetric mode:

$$x_1 = x_2 \tag{42}$$

   with frequency Ω_1.

2. Antisymmetric mode:

$$x_1 = -x_2 \tag{43}$$

with frequency Ω_2.

Both oscillations decay exponentially because of the damping factor

$$e^{-\gamma t} \tag{44}$$

3.1.2 Energy of the System

The total kinetic energy of the system is

$$T = \frac{1}{2}M\dot{x}_1^2 + \frac{1}{2}M\dot{x}_2^2 \tag{45}$$

Using the normal coordinates

$$x_1 = \frac{u+v}{2}, \qquad x_2 = \frac{u-v}{2} \tag{46}$$

we obtain

$$\dot{x}_1 = \frac{\dot{u}+\dot{v}}{2}, \qquad \dot{x}_2 = \frac{\dot{u}-\dot{v}}{2} \tag{47}$$

Substituting into the kinetic energy expression,

$$T = \frac{1}{2}M\left(\frac{\dot{u}+\dot{v}}{2}\right)^2 + \frac{1}{2}M\left(\frac{\dot{u}-\dot{v}}{2}\right)^2 \tag{48}$$

Expanding and simplifying gives

$$T = \frac{M}{4}\left(\dot{u}^2 + \dot{v}^2\right) \tag{49}$$

The total potential energy is

$$V = \frac{1}{2}\kappa x_1^2 + \frac{1}{2}\kappa x_2^2 + \frac{1}{2}\kappa_{12}(x_1 - x_2)^2 \tag{50}$$

Since

$$x_1 - x_2 = v \tag{51}$$

and

$$x_1^2 + x_2^2 = \frac{1}{2}\left(u^2 + v^2\right) \tag{52}$$

the potential energy becomes

$$V = \frac{\kappa}{4}\left(u^2 + v^2\right) + \frac{1}{2}\kappa_{12}v^2 \tag{53}$$

Combining terms,

$$V = \frac{\kappa}{4}u^2 + \frac{\kappa + 2\kappa_{12}}{4}v^2 \tag{54}$$

The total mechanical energy is therefore

$$E = T + V \tag{55}$$

Substituting the underdamped solutions,

$$u(t) = e^{-\gamma t}\left[A\cos(\Omega_1 t) + B\sin(\Omega_1 t)\right] \tag{56}$$

$$v(t) = e^{-\gamma t}\left[C\cos(\Omega_2 t) + D\sin(\Omega_2 t)\right] \tag{57}$$

shows that both u^2, v^2, $\dot{u}^2$, and $\dot{v}^2$ contain the factor

$$e^{-2\gamma t} \tag{58}$$

Hence the total energy decays exponentially:

$$E(t) \propto e^{-2\gamma t} \tag{59}$$

3.2 Numerical Solution

3.2.1 Euler Method

Consider the differential equation

$$\ddot{v} - 10v + 0.5\dot{v} = 0$$

where $M = 1$, $k = 30$, $k_{12} = 20$, $b = 0.5$, $x_{10} = 1$, $x_{20} = -0.5$, and $\dot{x_{1o}} = \dot{x_{20}} = 0$
Rearranging:

$$\ddot{v} = 10v - 0.5\dot{v}$$

Let

$$r = \dot{v}$$

Then the equation becomes the system

$$\dot{v} = r$$

$$\dot{r} = 10v - 0.5r$$

with initial conditions

$$v(0) = 1.5, \qquad r(0) = 0$$

and step size

$$h = 0.1$$

The Euler formulas are

$$v_{k+1} = v_k + h\, r_k$$

$$r_{k+1} = r_k + h(10v_k - 0.5r_k)$$

Euler Method Results:

k	t_k	v_k	$r_k = \dot{v}_k$
0	0.0	1.5000	0.0000
1	0.1	1.5000	1.5000
2	0.2	1.6500	2.9250
3	0.3	1.9425	4.42875

As we derived analytical solution for this case:
The differential equation is

$$\ddot{v} + 0.5\dot{v} - 10v = 0$$

Assume a solution of the form

$$v = e^{rt}$$

Then the characteristic equation is

$$r^2 + 0.5r - 10 = 0$$

Using the quadratic formula:

$$r = \frac{-0.5 \pm \sqrt{0.25 + 40}}{2}$$

The roots are

$$r_1 \approx 2.9221, \qquad r_2 \approx -3.4221$$

Hence the analytical solution is

$$v(t) = C_1 e^{2.9221t} + C_2 e^{-3.4221t}$$

Using the initial conditions:

$$v(0) = 1.5$$

$$\dot{v}(0) = 0$$

we obtain

$$C_1 \approx 0.8083, \qquad C_2 \approx 0.6917$$

Thus,

$$v(t) \approx 0.8083 e^{2.9221t} + 0.6917 e^{-3.4221t}$$

Comparison Between Numerical and Analytical Solutions

t	$Euler$v	$Analytical$v
0.0	1.5000	1.5000
0.1	1.5000	1.5733
0.2	1.6500	1.7996
0.3	1.9425	2.1958

Error Calculation

The absolute error is calculated using

$$Error = |v_{exact} - v_{Euler}|$$

The comparison table with the error values is

t	$EulerMethod$	$AnalyticalSolution$	$AbsoluteError$
0.0	1.5000	1.5000	0.0000
0.1	1.5000	1.5733	0.0733
0.2	1.6500	1.7996	0.1496
0.3	1.9425	2.1958	0.2533

Example at $t = 0.1$:

$$Error = |1.5733 - 1.5000|$$

$$Error = 0.0733$$

Consider the other differential equation

$$\ddot{u} + 30u + 0.5\dot{u} = 0$$

Rearranging:

$$\ddot{u} = -30u - 0.5\dot{u}$$

Let

$$w = \dot{u}$$

Then the equation becomes the system

$$\dot{u} = w$$

$$\dot{w} = -30u - 0.5w$$

with initial conditions

$$u(0) = 1.5, \qquad w(0) = 0$$

and step size

$$h = 0.1$$

The Euler formulas are

$$u_{k+1} = u_k + h\, w_k$$

$$w_{k+1} = w_k + h(-30u_k - 0.5w_k)$$

Euler Method Results:

k	t_k	u_k	$w_k = \dot{u}_k$
0	0.0	1.5000	0.0000
1	0.1	1.5000	-4.5000
2	0.2	1.0500	-8.7750
3	0.3	0.1725	-11.48875

As we derived the analytical solution in this case:

$$\ddot{u} + 0.5\dot{u} + 30u = 0$$

Using Constant Coefficients Method, Assume a solution of the form

$$u = e^{rt}$$

Then the characteristic equation is

$$r^2 + 0.5r + 30 = 0$$

Using the quadratic formula:

$$r = \frac{-0.5 \pm \sqrt{0.25 - 120}}{2}$$

$$r = -0.25 \pm 5.4715i$$

Hence the analytical solution is

$$u(t) = e^{-0.25t} \left(C_1 \cos(5.4715t) + C_2 \sin(5.4715t)\right)$$

Using the initial conditions:

$$u(0) = 1.5, \qquad \dot{u}(0) = 0$$

we obtain

$$C_1 = 1.5, \qquad C_2 \approx 0.0685$$

Thus,

$$u(t) = e^{-0.25t} \left(1.5 \cos(5.4715t) + 0.0685 \sin(5.4715t)\right)$$

Comparison Between Numerical and Analytical Solutions

t	$EulerMethod$	$AnalyticalSolution$	$AbsoluteError$
0.0	1.5000	1.5000	0.0000
0.1	1.5000	1.2848	0.2152
0.2	1.0500	0.7199	0.3301
0.3	0.1725	−0.0168	0.1893

Error Calculation

The absolute error is calculated using

$$Error = |u_{exact} - u_{Euler}|$$

The comparison table with the error values is

t	$EulerMethod$	$AnalyticalSolution$	$AbsoluteError$
0.0	1.5000	1.5000	0.0000
0.1	1.5000	1.2848	0.2152
0.2	1.0500	0.7199	0.3301
0.3	0.1725	−0.0168	0.1893

Example at $t = 0.2$:

$$Error = |0.7199 - 1.0500|$$

$$Error = 0.3301$$

3.2.2 Python Code

The code starts by importing the required Python libraries: `numpy` for numerical calculations, `pandas` for organizing results in a table format, and `matplotlib.pyplot` for visualization. These tools are essential for solving and analyzing the system of ordinary differential equations (ODEs) numerically.

The physical parameters of the system are defined, including the masses m_1, m_2, spring constants k_1, k_2, k_3, and damping coefficients c_1, c_2. These parameters fully describe the physical behavior of the coupled mass-spring system and determine how the system evolves over time.

The initial positions and velocities of both masses are defined here. These values represent the system's initial state at time $t = 0$. Since ODE problems require initial conditions, these conditions are necessary to generate a unique numerical solution.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

m1 = 1.0;m2 = 1.5
k1 = 20;k2 = 30;k3 = 20
c1 = 0.5;c2 = 0.5
x1_0 = 1.0;v1_0 = 0.0
x2_0 = -0.5;v2_0 = 0.0
```

The system is modeled as a set of first-order ordinary differential equations (ODEs), where the original second-order equations of motion are rewritten using the state vector $Y = [x_1, v_1, x_2, v_2]$.

This formulation allows the system to be expressed in terms of velocities and accelerations derived from Newton's second law, Hooke's law, and damping forces. The time domain is discretized over the interval $t = 0$ to $t = 20$ seconds using a small time step $dt = 0.01$, creating a numerical grid required for solving the system since an analytical solution is not available.

```
def f(Y):
    x1, v1, x2, v2 = Y
    dx1 = v1
    dv1 = (-k1*x1 + k2*(x2 - x1) - c1*v1) / m1

    dx2 = v2
    dv2 = (-k2*(x2 - x1) - k3*x2 - c2*v2) / m2

    return np.array([dx1, dv1, dx2, dv2])
t0 = 0
tf = 20
dt = 0.01

t_values = np.arange(t0, tf, dt)

Y = np.array([x1_0, v1_0, x2_0, v2_0])

results = []
```

The system of ordinary differential equations is solved numerically using the fourth-order Runge–Kutta (RK4) method. At each time step, four intermediate slopes (k_1, k_2, k_3, k_4) are calculated to estimate the next state of the system with high accuracy and stability. During the simulation, the computed values of time, displacements, and velocities are stored and later converted into NumPy arrays for efficient processing. The results are then separated into individual variables for positions and velocities, preparing the data for energy calculations and graphical visualization.

```
for t in t_values:

    k1_ = f(Y)
    k2_ = f(Y + 0.5 * dt * k1_)
    k3_ = f(Y + 0.5 * dt * k2_)
    k4_ = f(Y + dt * k3_)

    Y = Y + (dt / 6) * (k1_ + 2*k2_ + 2*k3_ + k4_)

    results.append([t, Y[0], Y[1], Y[2], Y[3]])

results = np.array(results)

t = results[:,0]
x1 = results[:,1]
v1 = results[:,2]
x2 = results[:,3]
v2 = results[:,4]
```

The kinetic and potential energies of the system are calculated to determine the total mechanical energy and analyze the effect of damping on energy dissipation over time.

A `Pandas DataFrame` is used to organize the numerical results, including time, displacement, velocity, and total energy values, into a clear tabular format.

The displacement plot illustrates the coupled oscillatory motion of the two masses and the gradual reduction in amplitude caused by damping.

The energy plot demonstrates the continuous decrease of total mechanical energy over time due to damping effects in the non-conservative system.

```
KE1 = 0.5 * m1 * v1**2
KE2 = 0.5 * m2 * v2**2

PE1 = 0.5 * k1 * x1**2
PE2 = 0.5 * k2 * (x1 - x2)**2
PE3 = 0.5 * k3 * x2**2

E_total = KE1 + KE2 + PE1 + PE2 + PE3

df = pd.DataFrame({
    "Time": t,
    "x1": x1,
    "v1": v1,
    "x2": x2,
    "v2": v2,
    "Energy": E_total})

print(df.head(20))
```

4 Discussion

The numerical results demonstrate the coupled oscillatory behavior of the two masses and the continuous energy exchange between them through the connecting spring. Due to the damping effect, the oscillation amplitudes and total mechanical energy gradually decrease over time until the system approaches equilibrium.

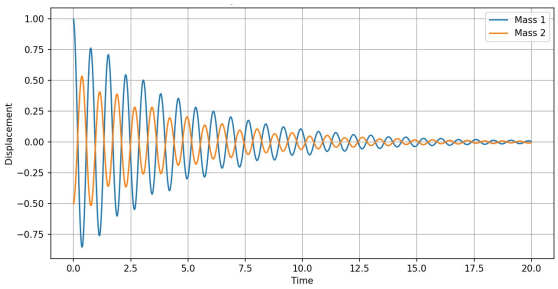


Figure 1: Coupled Oscillator(RK4 solution)

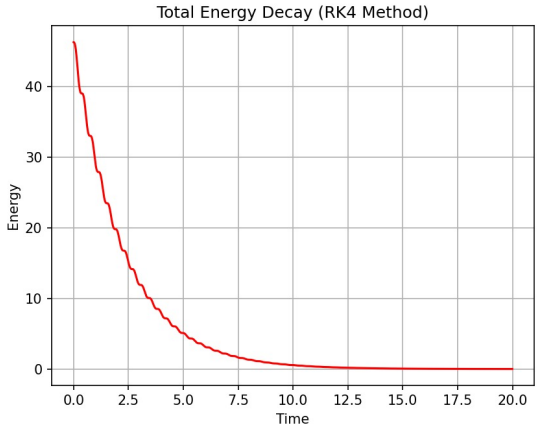


Figure 2: Energy Decay

The dynamic behavior of the damped two-mass, three-spring system is strongly influenced by the physical parameters: the masses, spring constants, and damping coefficients. Each parameter affects the oscillation frequencies, amplitude decay, and energy transfer between the two masses.

4.1 Effect of Masses (m_1, m_2)

The masses determine the inertia of the system. Increasing either mass results in slower acceleration and therefore:

- Lower natural frequencies, since the system becomes more resistant to motion.
- Longer oscillation periods, as the system takes more time to complete each cycle.
- Reduced sensitivity to coupling effects when one mass is significantly larger than the other, leading to asymmetric motion.

If $m_1 = m_2$, the system becomes symmetric, which simplifies structure and makes energy exchange between masses more balanced.

4.2 Effect of Spring Constants (k_1, k_2, k_3)

The spring constants control the stiffness of the system, directly affecting the restoring forces.

- Increasing any spring constant leads to higher natural frequencies due to stronger restoring forces.
- The central spring k_2 governs the coupling strength:
  - Large k_2: strong coupling and greater frequency
  - Small k_2: weak coupling and nearly independent motion of the masses.
- The outer springs k_1 and k_3 mainly control the individual oscillatory behavior of each mass.

4.3 Effect of Damping Coefficients

Damping controls the rate of energy dissipation in the system.

- Increasing damping leads to faster decay of oscillations.
- Amplitude decreases more rapidly over time.
- The system may transition from underdamped to critically damped or overdamped behavior.

In the underdamped case, oscillations persist but with an exponentially decaying envelope. Damping also reduces the observed oscillation frequencies compared to the undamped case.

Overall, the damped two-mass, three-spring system demonstrates how coupling and damping significantly influence vibrational behavior. Instead of simple harmonic motion, the system exhibits decaying amplitudes, showing how energy is redistributed and gradually lost over time. This model is widely applicable in mechanical systems, molecular vibrations, and structural dynamics.

4.4 Summary

Overall, the system response is highly sensitive to parameter variation. Masses control inertia, spring constants determine stiffness and coupling strength, and damping governs energy dissipation. The interplay between these parameters defines whether the system exhibits persistent coupled oscillations or rapidly decaying motion toward equilibrium.

5 References

References

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J. B. Marion and S. T. Thornton, *Classical Dynamics of Particles and Systems*, 5th ed., Brooks Cole, 2003.